

Project Title

Automated Network Request Management in ServiceNow

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Project overview The Automated Network Request Management in ServiceNow project focuses on simplifying and accelerating the handling of network-related service requests. By leveraging ServiceNow's service catalog, workflows, and automation capabilities, the solution eliminates manual bottlenecks and ensures requests are processed efficiently. End users can easily submit requests through a self-service portal, while automated routing, approvals, and task assignments streamline fulfillment. Where possible, integration with network automation tools enables end-to-end execution without human intervention. This results in faster turnaround, improved SLA compliance, and enhanced transparency for both users and IT teams.

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Introduction

Modern enterprises rely heavily on robust and efficient network services to support day-to-day business operations. As organizations grow, the demand for network-related requests—such as access provisioning, configuration changes, and connectivity support—also increases. Traditional manual request handling often leads to delays, errors, and limited visibility, impacting both productivity and user satisfaction. To address these challenges, the Automated Network Request Management in ServiceNow project introduces a streamlined solution that leverages ServiceNow's workflow engine, service catalog, and automation features. By enabling end users to submit requests through a self-service portal and automating approvals, task assignments, and notifications, the system ensures faster, more accurate, and transparent request management. This initiative not only enhances the user experience but also empowers IT teams by reduci

ng repetitive tasks, improving SLA compliance, and allowing focus on strategic, high-value activities.

Project Objective

This project aims to design and implement a streamlined, automated solution for managing network related service requests within ServiceNow. It enables end users to submit requests for network services through a user-friendly self-service portal. The system leverages ServiceNow's workflow engine, catalog items, and approval processes to ensure requests are properly captured, validated, and routed for fulfillment. Upon submission, requests trigger automated notifications, task assignments, and—where applicable—integration with network automation tools or scripts to fulfill standard requests without manual intervention.

Key Features

- Custom service catalog for common network requests
- Dynamic forms to capture relevant request details
 - Automated approval workflows based on request type and sensitivity
- Integration with infrastructure management or orchestration tools (optional)
- Realtime status updates and notifications to requesters and technicians Reporting and analytics on request volume, resolution time, and SLA adherence

ServiceNow Developer Setup:

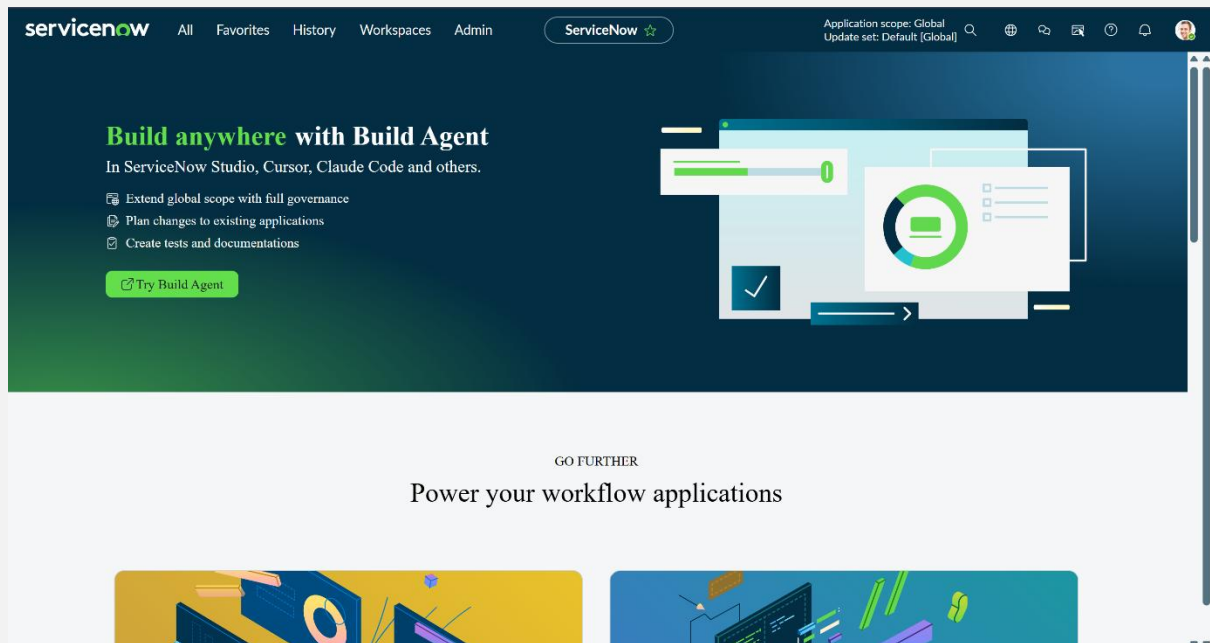
Create a Developer Account

1. Go to ServiceNow Developer Portal(<https://developer.servicenow.com/dev.do>).
2. Sign up for a free developer account and fill out the following details. .
3. After signing up, you will get a verification mail to your provided email id. After the verification your ServiceNow Developer Portal Home Page will appear.
4. Now click on start building it will take you to the section where you can request a Personal Developer Instance (PDI) or start using App Engine Studio and other tools.

Project Implementation in ServiceNow:

After the instance building is completed, the page will be directed to your creator studio . Creator Studio in ServiceNow provides a guided.

nocode environment to build applications quickly. It is especially useful for creating requestbased applications by defining forms, setting up tables, and automating workflows

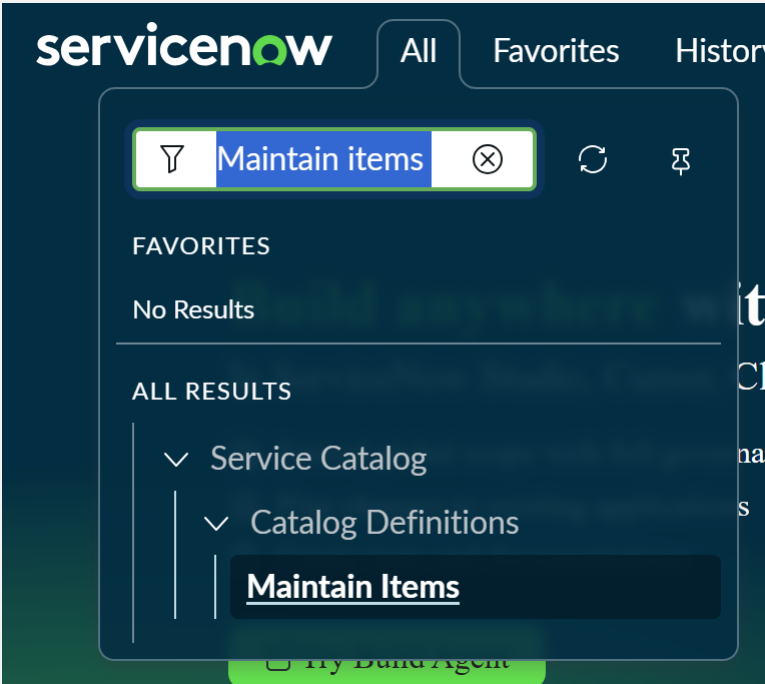


a. Service Catalog Creation

i. Creation of Service Catalog

1. Navigate to Application navigator
2. Click on All >> search for Service Catalog
3. Under Service Catalog>> Maintain items
4. Click on New
5. Fill the details >> Name– Network Request
6. Select Catalog>> Service Catalog
7. Select Category>> Network
8. Fill the Short Description as Network request Management

9. Click on Save.



Click on New

Catalog Items									
Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated	
3M Privacy Filter - Lenovo X1 Carbon	Privacy Filter - X1 Carbon	true		Service Catalog	Peripherals	\$43.19	Item	2022-11-20 20:46:33	
3M Privacy Filter - MacBook Pro	Privacy Filter	true		Service Catalog	Peripherals	\$42.23	Item	2022-11-20 20:46:33	
3M Privacy Filter - Macbook Pro Retina	Privacy Filter	true		Service Catalog	Peripherals	\$40.31	Item	2022-11-20 16:00:00	
Access	Microsoft Access	true		Service Catalog	Software	\$139.99	Item	2022-11-20 20:46:33	
Acrobat	Adobe Acrobat	true		Service Catalog	Software	\$139.99	Item	2022-12-05 20:46:33	
Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-07-19 21:21:30	
Add/Remove users from group	Add/Remove users from group	true		Service Catalog	Services	\$0.00	Item	2022-12-05 10:17:33	
Adobe Acrobat Pro	Create, edit or convert PDF files	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33	
Adobe Creative Cloud	More connected ways of creating and shar...	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33	
Apple iPad 3	Apple iPad 3	false		Service Catalog	Tablets	\$600.00	Item	2022-11-20 20:46:33	
Apple iPhone 13	Request for Apple iPhone 13	true		Service Catalog	Mobiles	\$799.00	Item	2024-08-08 02:16:16	
				Service Catalog				2024-08-08	

Create the Network Request

servicenow All Favorites History Workspaces Catalog Item - New Record ☆ Application scope: Global Update set: Default [Global]

Build and modify items faster with the improved [Catalog Builder](#).

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Network Request Application: Global

Catalogs: Service Catalog Active: ☒

Category: Network Standard Changes Fulfillment automation level: Unspecified

State: -- None --

Checked out: -- None --

Owner: System Administrator

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description: Network request Management

Description: <>

Rich text editor toolbar: B, I, U, Link, Unlink, Bold, Italic, Underline, Text color, Background color, Bulleted list, Numbered list, Indent, Outdent, Undo, Redo, Source code.

ii. Variables Configuration

Open the catalog item just created.

Scroll down to the Variables related list and click New to create form fields.

1. Select Variables type as Single, Multi line text, reference, choices etc as per requirement
2. Catalog item– Network Request
3. Order–100,200,300,,,
4. Question– provide the variable label
5. Name–provide the variables name(used for scripting)
6. Tooltip– this will appear when cursor overed on the field
7. Example text – this will suggest what we need to enter on the field.
8. Mandatory, Read-Only– need to configure on demand.
9. Auto populate– need to select dependent variable, apply dot walking to get selected value.
10. Click on Save or Submit.

Type	Question	Order
Container Start	Service Details	200
Multiple Choice	Is this a New connection or Relocation?	300
Single Line Text	If this is a relocation, Please provide...	310
Single Line Text	If this is a relocation, Please provide...	320
Container Start	Location and Device Type	400
Single Line Text	Please provide address here	410
Multiple Choice	Types of devices	420
Single Line Text	Provide device details here	430
Container Start	Additional Information	500
Single Line Text	If anything else, please specify	510

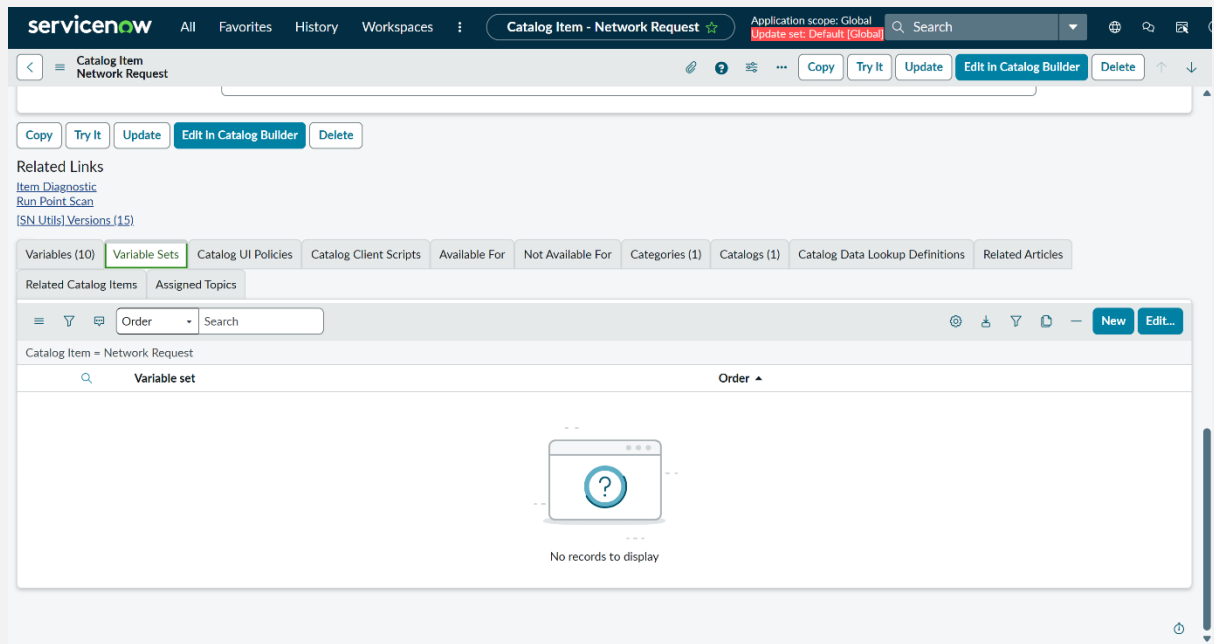
iii. Variables Types

1. Is this a New connection or Relocation? >> Choice >> New/ Relocation/None
2. If this is a relocation, Please provide your relocated address here>>String
3. Types of devices>> Choice>> Laptop/Mobiles/Others
4. Please provide address here>>String
5. Provide device details here>> String
6. If anything else, please specify>> String

iv. Variable Set Configuration

To enhance form usability:

- Navigate to the Variable Sets (optional).
- Follow the same procedure as we used for Variables Creation, for the variable set as well.
- Apply variable sets to the catalog item.



Variables Types

1. Opened on behalf of >> Reference>> reference to user table
2. Email Id >> Single line text >> Auto populate by Opened on behalf of variable.
3. User name >>Single line text >> Auto populate by Opened on behalf of variable.
- 4.Phone Number >>Single line text >> Auto populate by Opened on behalf of variable.
5. Proof of Document >> Attachment

The screenshot shows the ServiceNow interface for configuring a Variable Set named 'Request Information'. The top navigation bar includes 'All', 'Favorites', 'History', 'Admin', and a search bar. The main header indicates the 'Application scope: Global' and 'Update set: default (Global)'. The variable set configuration form includes fields for Title (Request Information), Internal name (request_information), Order (100), Type (Single Row), and Description. The Application is set to Global, and the Layout is 1 Column Wide. Below the form are 'Update' and 'Delete' buttons. A tabbed interface shows 'Variables (5)' selected, displaying a list of variables for the 'Request Information' set.

Name	Type	Question	Order
opened_on_behalf_of	Reference	Opened on behalf of	
proof_of_document	Attachment	Proof of Document	
email_id	Single Line Text	Email Id	
user_name	Single Line Text	User name	
phone_number	Single Line Text	Phone Number	

v. Catalog UI Policy Configuration

Scenario: If user selects types of devices is Others, then Please specify field should populate.

Procedure:

1. Navigate to catalog items
2. Open Network Request item
3. In related list, we have Catalog UI policy
4. Click on New button to configure New UI policy
5. Select Applies to as Catalog item
6. Select catalog item as Network Request
7. Provide short description, if required
8. Apply condition>> types of devices is others
9. Click on save, after saving the form will get UI policy actions in the related list
10. Click on New button to configure new UI Policy action, and Select the variable which we want to display on condition
11. Make Visible True as per our requirement
12. Update the UI Policy and Test the same on Catalog form

Catalog Item

Network Request

Press Alt+0 for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.

Meta

Copy

Try It

Update

Edit In Catalog Builder

Delete

Related Links

[Item Diagnostic](#)

[Run Point Scan](#)

[\[SN Utils\] Versions \(17\)](#)

Variables (10)

Variable Sets (1)

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

Variable set

Search

Actions on selected rows...

New

Edit...

Catalog Item = Network Request

Variable set	Order
Request Information	100

1 to 1 of 1

servicenow

All

Favorites

History

Admin

Variable Set - Request Information

Application scope: Global

Update set: Default (Global)

Variable Set

Request Information

Update

Delete

* Title

Request Information

* Internal name

request_information

Order

100

Type

Single Row

Description

Application

Global

Display title

Layout

1 Column Wide

Update

Delete

[SN Utils] Versions (6)

Variables (5)

Catalog UI Policies

Catalog Client Scripts

Included In (1)

Catalog Data Lookup Definitions

Order

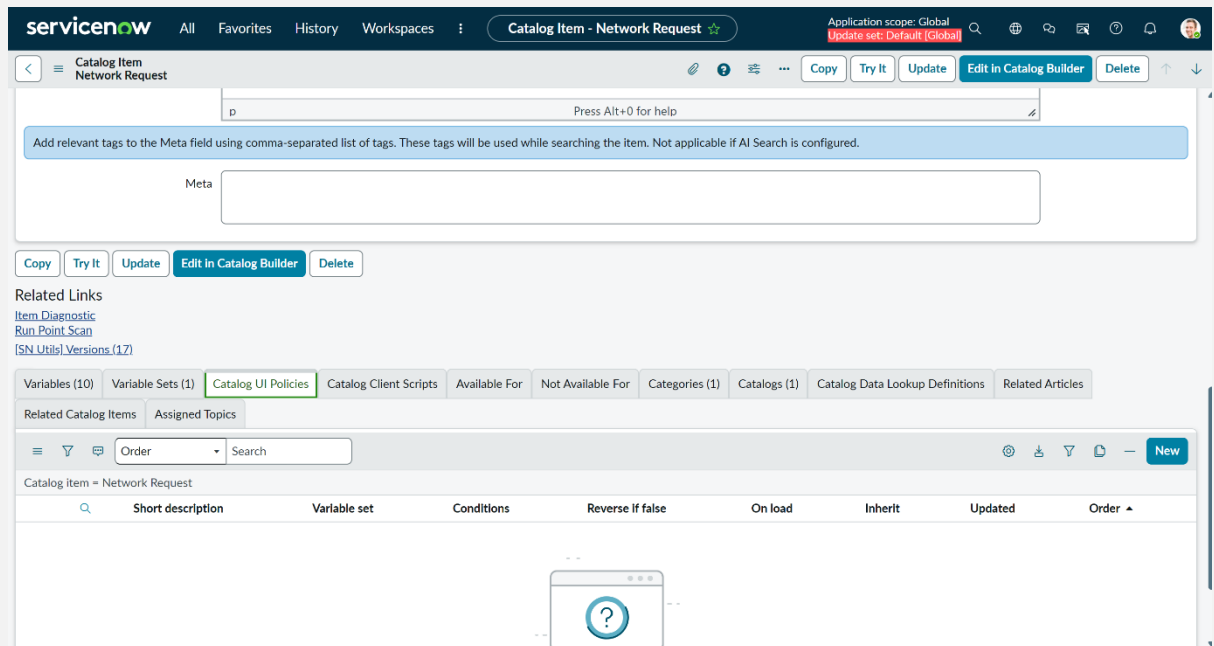
Search

Actions on selected rows...

New

Variable set = Request Information

Name	Type	Question	Order
opened_on_behalf_of	Reference	Opened on behalf of	
proof_of_document	Attachment	Proof of Document	
email_id	Single Line Text	Email Id	
user_name	Single Line Text	User name	
phone_number	Single Line Text	Phone Number	



b. Creation of Table

i. Creation of Table

1. Navigate to: System Definition > Tables.

a. Click New to create a new table.

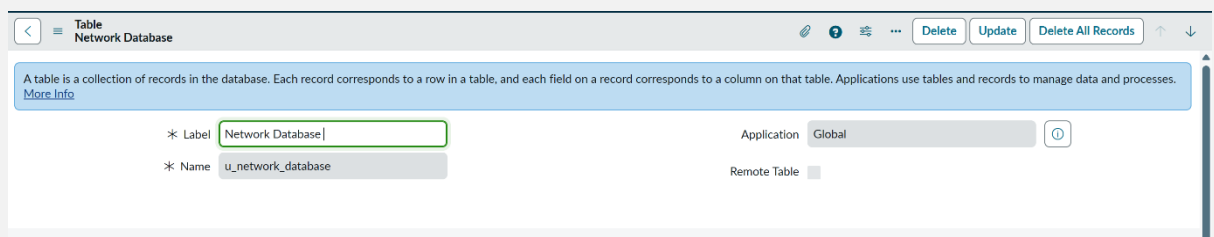
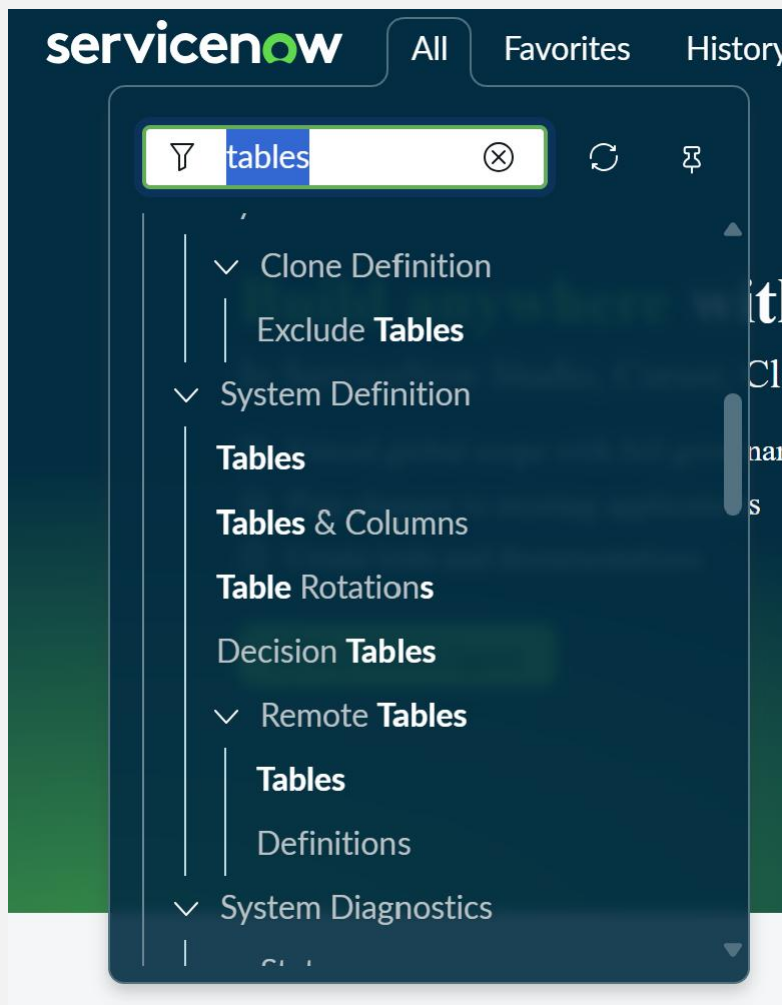
b. Fill in Table Information:

Name: Name of the table -----

Label: Backend name of the table-----

Autogenerate schema: Leave it checked if you'd like ServiceNow to auto-generate schema fields.

c. Click Submit to create the table.



ii. Creation of fields

In ServiceNow, fields are created at the table level. To create a field, you first need to identify the table where the field will reside.

1. In the Application Navigator (left-side panel), type Tables in the search bar.

2. Under System Definition, click Tables. This will take you to a list of all tables in the system.

a. Select the Table to Add the Field

From the list of tables, search for and select the table you want to add a field to.

For example, if you want to add a field to the Network database table:

1. Type "Network database" in the search box or scroll through the list.
2. Click on the Network database table name. You'll now see a list of all fields (columns) associated with the Network database table.

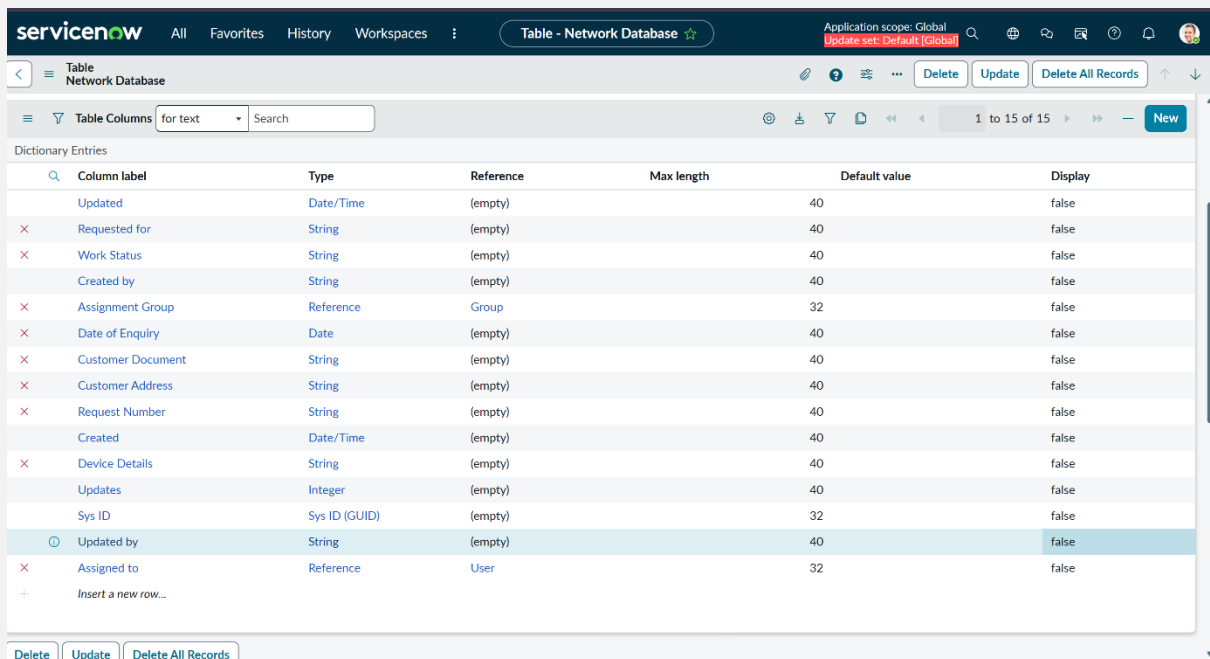
b. Open the Table's Columns

After selecting the table, you'll be brought to a view that lists all the columns (fields) that currently exist on that table.

To create a new field (column), go to the Columns tab (this is where all fields for the selected table are listed).

c. Create a New Field

1. In the Columns tab, click the New button located at the top-right corner of the page to create a new field.
2. You'll now be prompted with a form where you need to define the new field. The following fields need to be filled out:

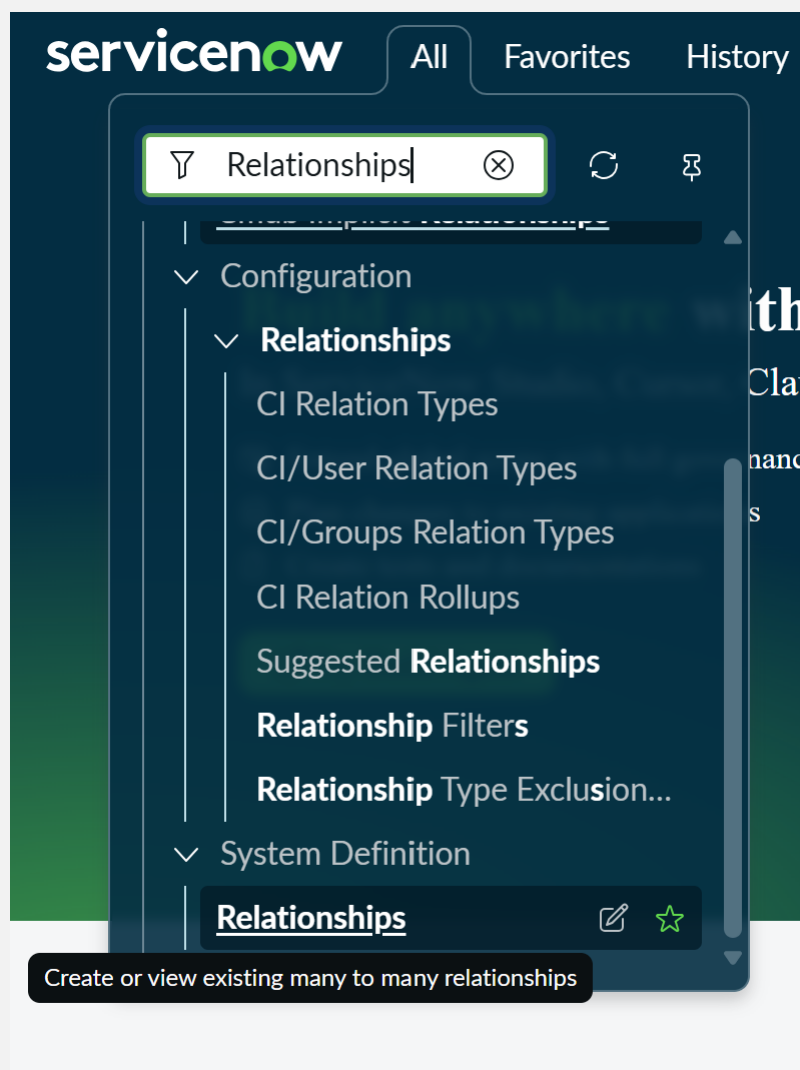


Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40		false
Requested for	String	(empty)	40		false
Work Status	String	(empty)	40		false
Created by	String	(empty)	40		false
Assignment Group	Reference	Group	32		false
Date of Enquiry	Date	(empty)	40		false
Customer Document	String	(empty)	40		false
Customer Address	String	(empty)	40		false
Request Number	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Device Details	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated by	String	(empty)	40		false
Assigned to	Reference	User	32		false
Insert a new row...					

c. Request Approvals Creation(Related List)

i. Creation of Related List

1. Navigate to System Definition > Relationships.
2. Click New to create a new relationship.
3. Fill in the following details:
 - o Name: Approval Request
 - o Applies to Table : Network Database table.
 - o Queries from Table : Sysapprovals table.
 - o Active: Make sure it's set to True.
3. Save the relationship.



servicenow All Favorites History Workspaces : Relationship - New Record ☆ Application scope: Global Update set: Default [Global]

< Relationship New record

Name

Advanced ☐

Applies to table Network Database [u_network_datab...]

Queries from table Approval [sysapproval_approver]

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see [the documentation](#) See also the article about the [recommended form of the script](#).

Query with <> Turn on ECMAScript 2021 (ES12) mode

```

1 (function refineQuery(current, parent) {
2
3   // Add your code here, such as current.addQuery(field, value);
4
5 })(current, parent);

```

Submit

ii. Adding Related List to the Table

1. Navigate to Form Designer for the table where you want to show related records.
2. Add a Related List widget to the form.
3. Select the Related List you want to show

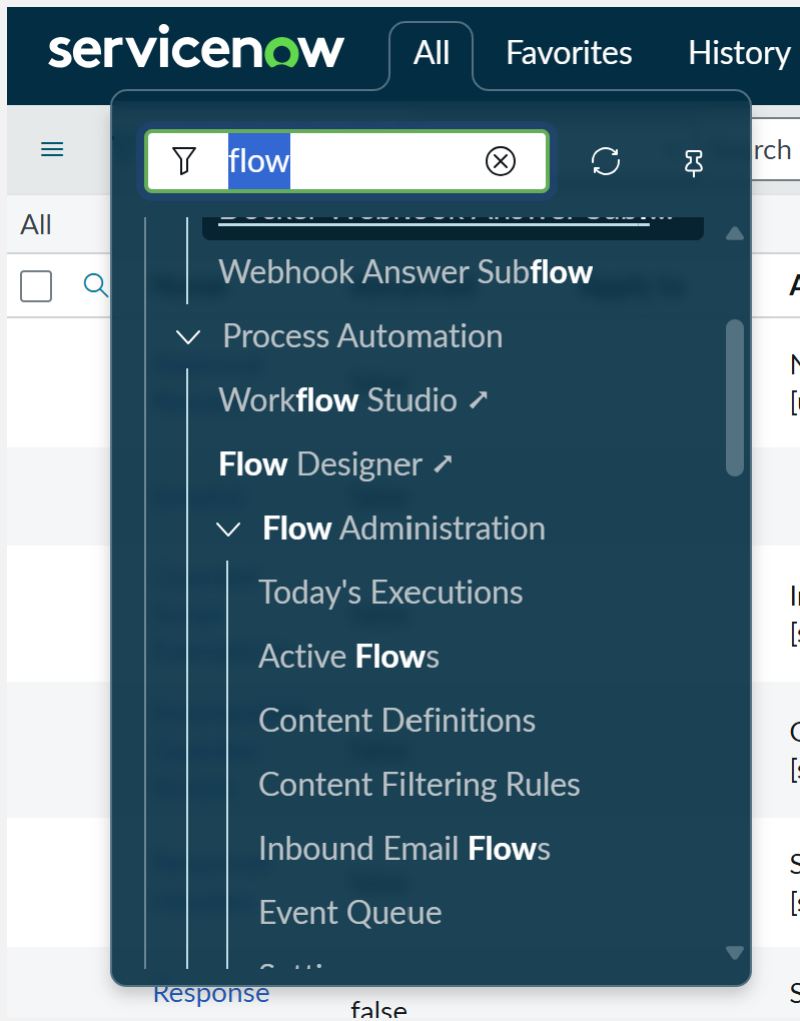
d. Creation & Implementation of flows, Actions in Flow Designer

i. Creation of Flow

1. Navigate to Flow designer home page
2. Click on New to create a new flow
3. Provide flow name as Network Request
4. Provide description of flow
5. Click on Build flow.

ii. Configuring Trigger

1. Click on (+) Icon to Configure the Trigger
2. Select Trigger as Application >> Service catalog
3. Click on Done.



Workflow Studio

Homepage Operations Integrations

Playbooks Flows Subflows Triggers Actions Decision tables

Flows 93

Last refreshed just now.

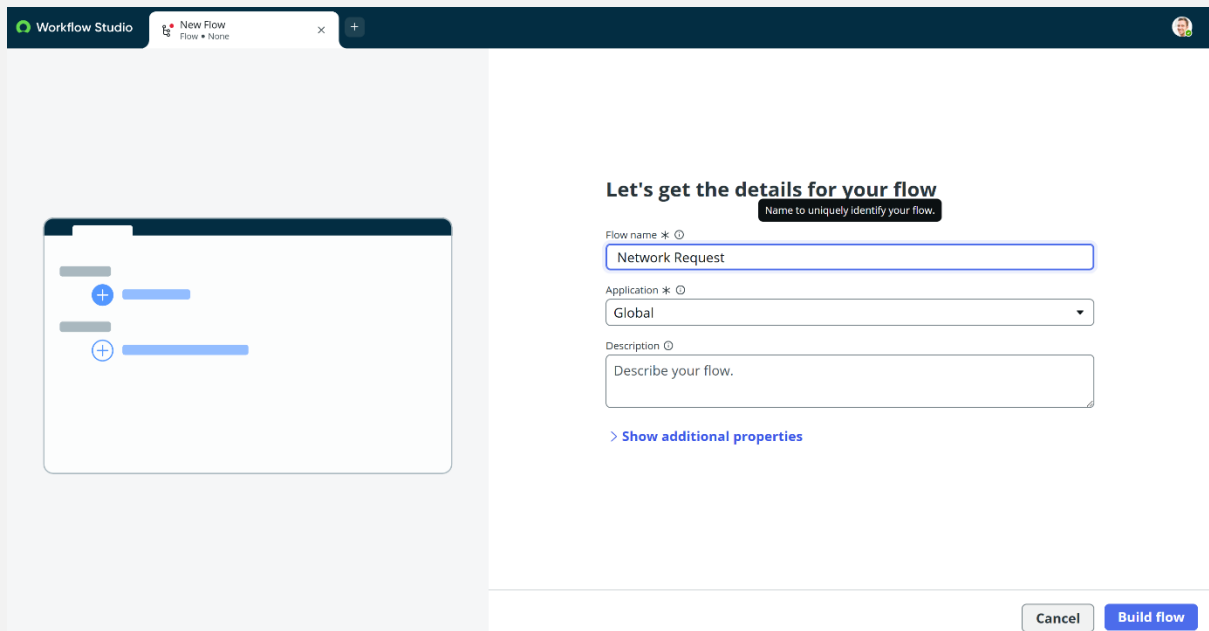
Name	Application	Status	Active	Updated
Application Intake Request Flow (Deprecated)	Application Intake	Published	false	2026-07-19 21:42:52
Application Intake Request V2	Application Intake	Published	true	2026-07-19 21:42:46
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-02-02 20:07:22
Business process approval flow	Global	Published	true	2020-09-27 22:06:13
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05
Change - Conflict Detection	Global	Published	true	2026-02-02 19:59:23
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26
Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08
Change - Normal - Assess	Global	Published	true	2020-10-06 05:37:05
Change - Normal - Authorize	Global	Published	true	2020-10-06 05:38:35

Pick up where you left off

- Deployment Migration t...
Last updated: a year ago by Syste...
- Multi-factor Authenticati...
Last updated: a year ago by Syste...
- Steps
Last updated: a year ago by Syste...

Latest updates

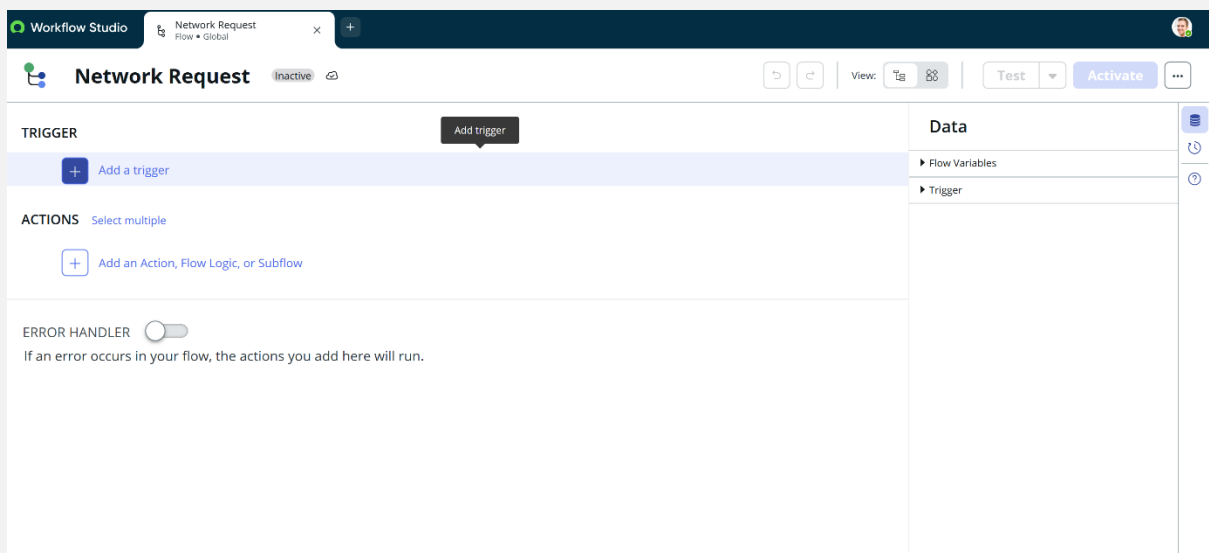
- System Administrator modified Deployment Migration to ReleaseOps
a year ago
- System Administrator modified Multi-factor Authentication
a year ago
- System Administrator modified Steps
a year ago
- System Administrator modified On-Call Scheduling
a year ago
- System Administrator modified Display FIDO2 as an MFA Factor
a year ago



The screenshot shows the 'New Flow' dialog in Workflow Studio. On the left is a canvas with two placeholder blocks. On the right, the 'Let's get the details for your flow' section contains the following fields:

- Flow name ***: A text input field containing 'Network Request'. A tooltip above it says 'Name to uniquely identify your flow.'
- Application ***: A dropdown menu with 'Global' selected.
- Description**: A text area with the placeholder text 'Describe your flow.'

Below these fields is a link '> Show additional properties'. At the bottom right are 'Cancel' and 'Build flow' buttons.



The screenshot shows the configuration page for a 'Network Request' flow. The top bar includes the title 'Network Request', an 'Inactive' status, and buttons for 'Test' and 'Activate'. The main area is divided into three sections:

- TRIGGER**: Contains a '+ Add a trigger' button and an 'Add trigger' tooltip.
- ACTIONS**: Contains a '+ Add an Action, Flow Logic, or Subflow' button and a 'Select multiple' link.
- ERROR HANDLER**: Includes a toggle switch and the text 'If an error occurs in your flow, the actions you add here will run.'

On the right side, there is a 'Data' panel with a list of variables: 'Flow Variables' and 'Trigger'.

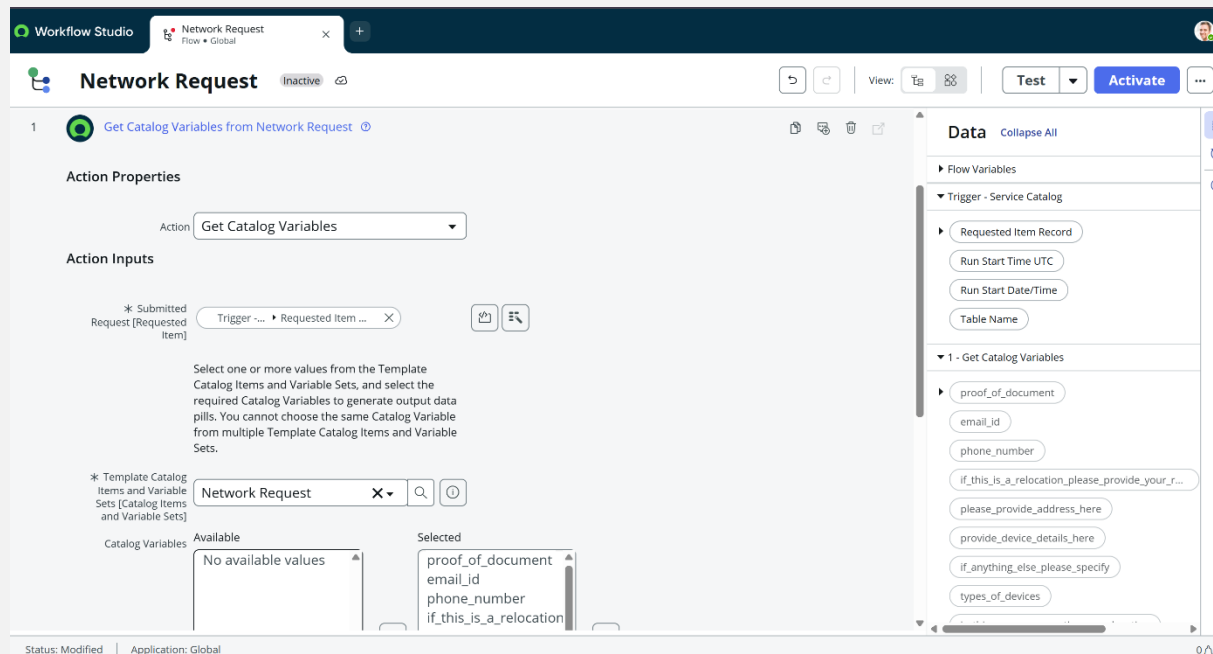
iii. Configuring Actions

Click on Actions button to configure new action

1. Get Catalog Variables

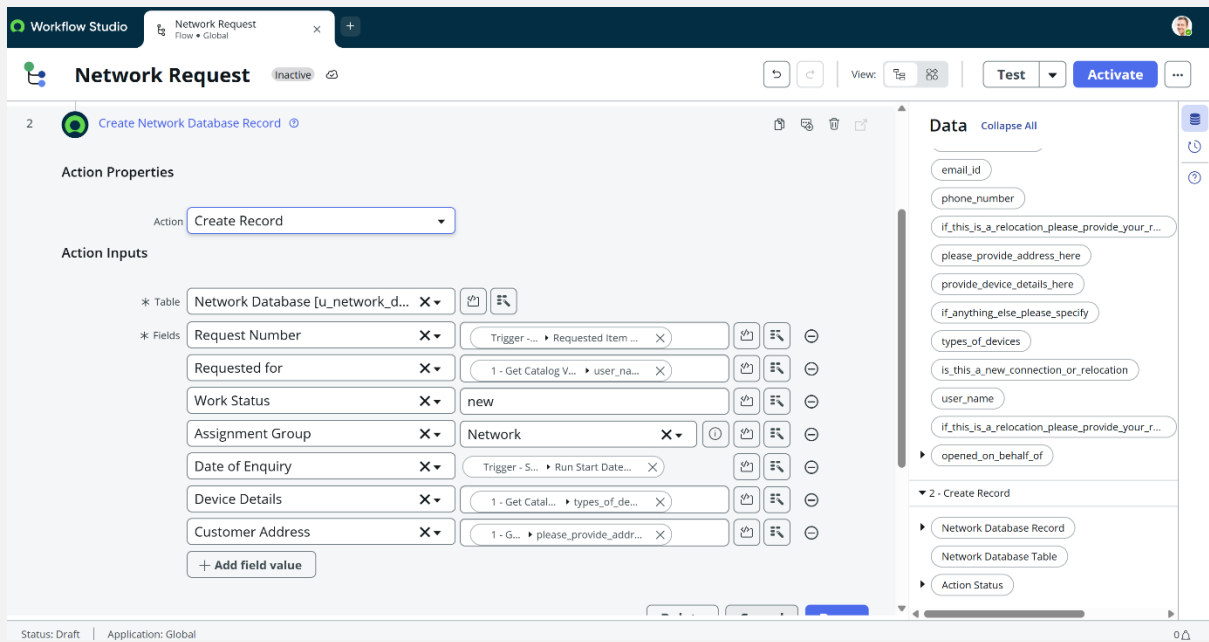
- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs>> Trigger>>service catalog>>Requested Item
- Template catalog items >> Select table >> Network Request

- Select the Required Variables and Move to the selected area.
- Click on done



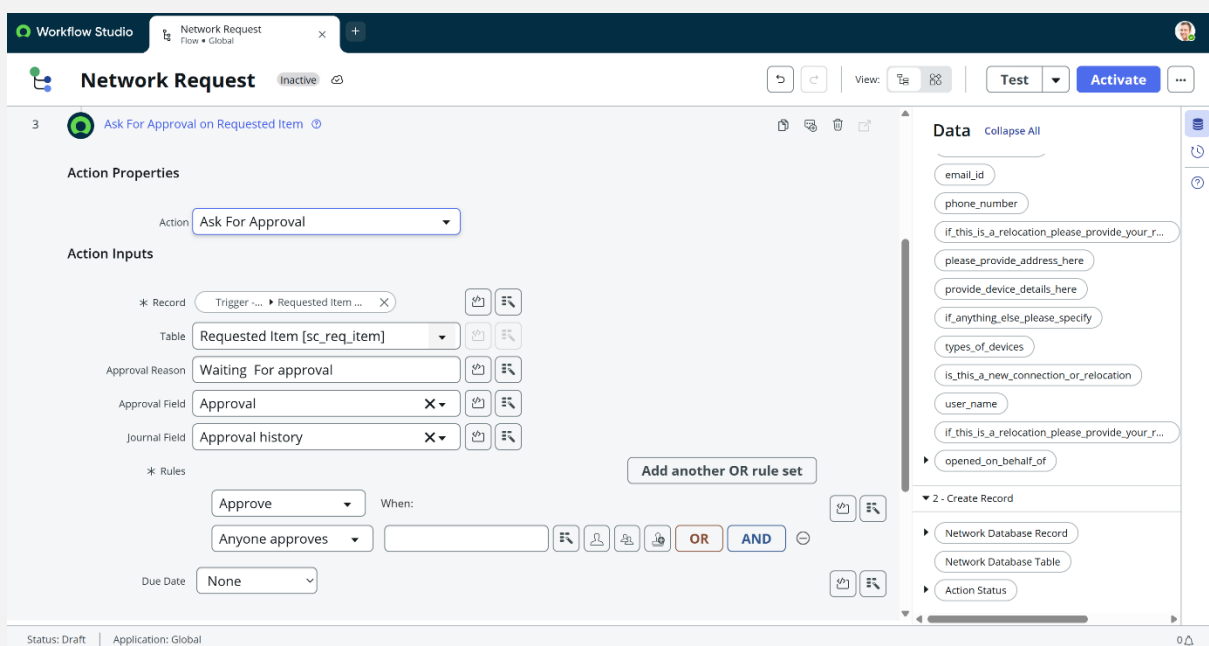
2. Create Record

- Select action as Create Record
- Select table as Network Database
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done



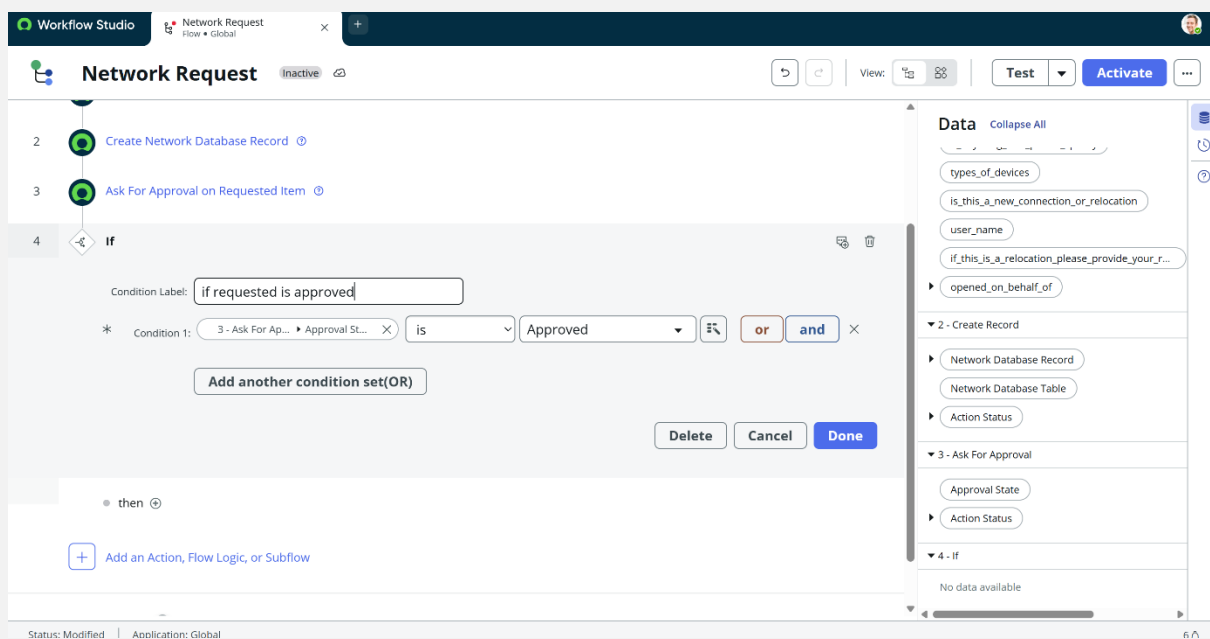
3. Ask for approvals

- Select action as Ask for Approval
- Select target record >> Create record>> network database table
- Provide Approval Reason>> Waiting for approval
- Configure approval rules>> Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below
- Click on done



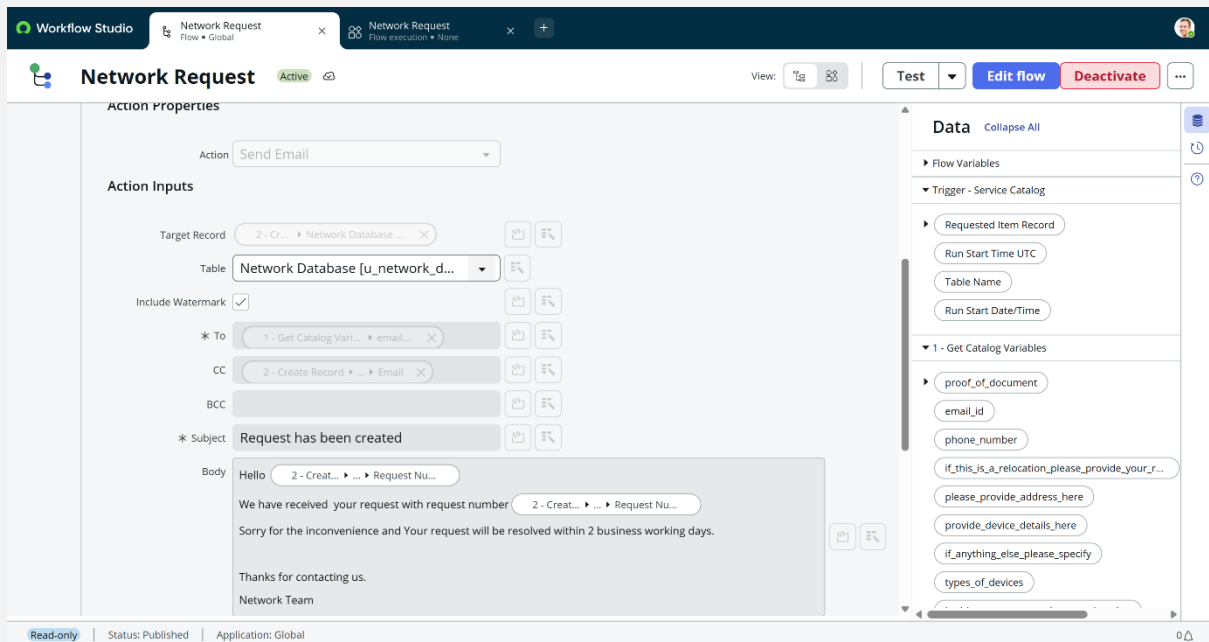
4. Flow Logic

- Select action as flow logic and Select If condition
- Apply condition >> Ask for approvals state is Approved/Rejected as per requirement
- Click on done



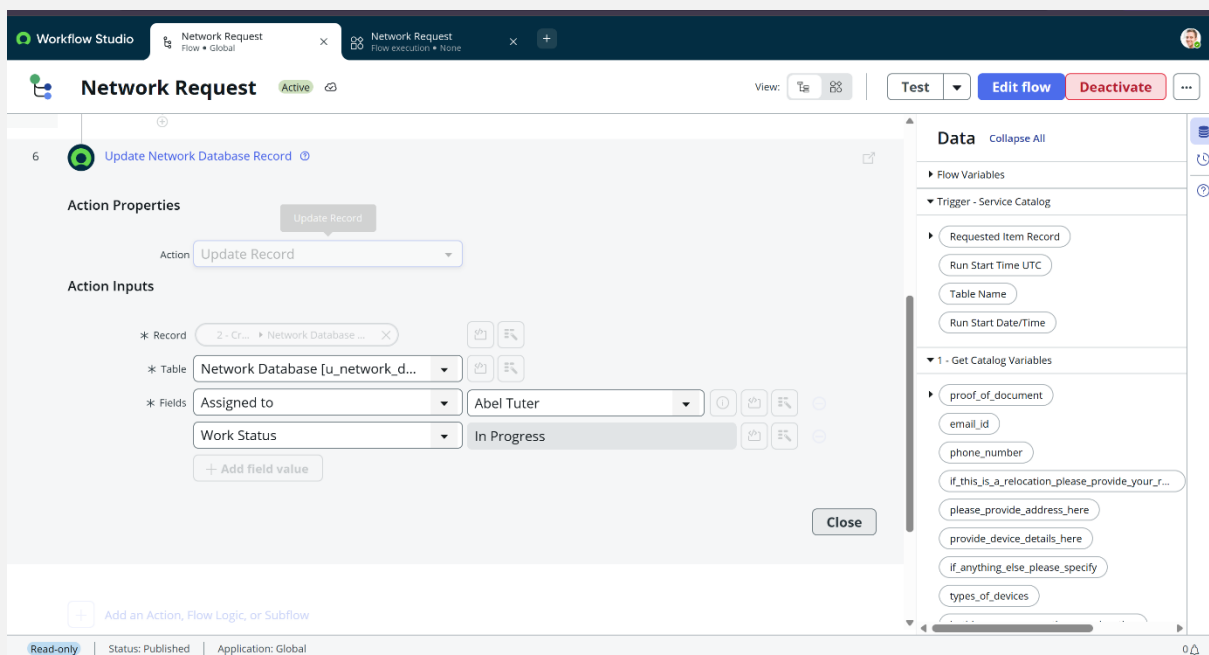
5. Send Email

- Select action as Send Email
- Select target record >> Create record>> network database table
- Table will be selected automatically
- Configure To, CC, BCC as per our requirements(select static/dynamic)
- Provide Subject & Body as shown in the below picture
- Click on done



6. Update Record

- Select action as Update Record
- Select record as >> create record>> network database
- Table will be selected automatically
- Configure the fields as per requirement, as shown in below
- Click on done



iv. Flow Chart

FLOW STATISTICS

TRIGGER

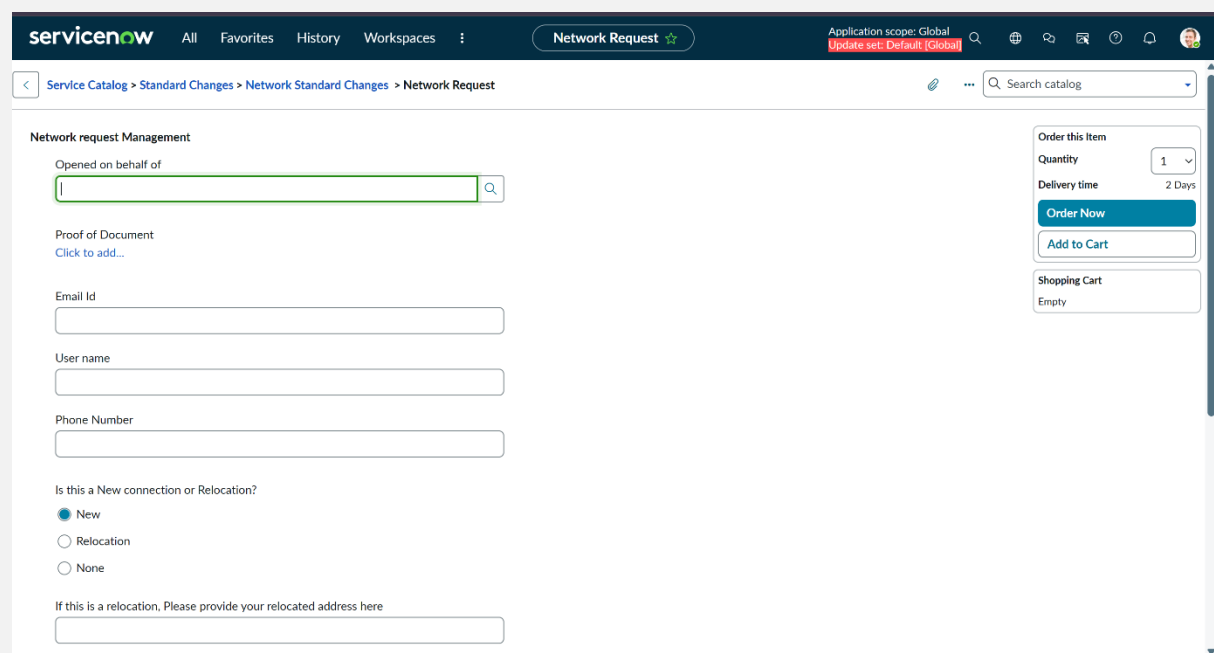


ACTIONS

- 1  **Get Catalog Variables** from Network Request
- 2  **Create Record**
- 3  **Ask For Approval**
- 4  **If if requested is approved**
- 5  **Send Email**
- 6  **Update Record**

ERROR HANDLER

V.Screenshots of Output



The screenshot shows the ServiceNow 'Network Request' form. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a 'Network Request' button. The breadcrumb trail is 'Service Catalog > Standard Changes > Network Standard Changes > Network Request'. The main form area is titled 'Network request Management' and contains several input fields: 'Opened on behalf of' (with a search icon), 'Proof of Document' (with a 'Click to add...' link), 'Email Id', 'User name', and 'Phone Number'. Below these is a radio button group for 'Is this a New connection or Relocation?' with options 'New' (selected), 'Relocation', and 'None'. At the bottom, there is a text field for 'If this is a relocation, Please provide your relocated address here'. On the right side, there is a 'Shopping Cart' section with 'Order this Item' (Quantity: 1, Delivery time: 2 Days), 'Order Now' and 'Add to Cart' buttons, and an 'Empty' shopping cart.

VI.EXECUTION DETAILS

Workflow Studio

Network Request

Network Request

Network Request

Network Request

Network Request

Network Request

Network Request

EXECUTION DETAILS

Network Request

Test Run - Completed

Open flow

Open context record

Show Action Details

State

Start time

FLOW STATISTICS

Run as: System Administrator

Open flow logs

Completed

2026-08-11 08:32:03

1945ms

TRIGGER

Catalog Item Requested

ACTIONS

1

Get Catalog Variables from Network Request

Core Action

Completed

2026-08-11 08:32:03

29ms

2

Create Record

Core Action

Completed

2026-08-11 08:32:03

11ms

3

Ask For Approval

Core Action

Completed

2026-08-11 08:32:03

1125ms

4

If if requested is approved

Flow Logic

Evaluated - True

2026-08-11 08:33:15

768ms

5

Send Email

Completed

2026-08-11 08:33:15

768ms

6

Update Record

Core Action

Completed

2026-08-11 08:33:16

8ms

ERROR HANDLER

VII. EMAIL

servicenow

Close

We received your request

Hi System,

We created **REQ0010005** to handle your recent request.

You can view your request to track updates and make changes.

View request

About this request

Requested item number: **RITM0010005**

Short description: **Network Request**

Quantity: 1

VIII. Conclusion

The Automated Network Request Management in ServiceNow project provides a streamlined, reliable, and

user-friendly solution for handling network-related service requests. By automating approvals, routing, and

notifications, the system eliminates delays and errors associated with manual processes while offering

transparency through real-time updates and reporting.

This initiative not only improves SLA compliance and user satisfaction but also optimizes IT operations

by reducing repetitive workload and enabling faster fulfillment. Ultimately, it supports the organization's

goal of delivering efficient, high-quality, and responsive network services, empowering both end

users and IT teams.